

Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title:

Future of the fabric

Team Name:

Style Gen

Team Members:

- Sk Ameerunnisa
- P Hemalatha
- S Sirisha
- T Ramya
- V Mounika

Phase-1: Brainstorming & Ideation

Objective:

Customizing the clothes according to the wish of the customer by using the stable diffusion pipeline

Purpose of this is to get quality clothes for customer and user friendly relation between the designer and customer

Impact of this is not to waste time, money and interest in clothing

Key Points:

1. **Problem Statement:** Design and develop a clothing image generator using the stable diffusion pipeline that can produce high quality, photorealistic images of clothing items based on text prompts.
 2. **Proposed Solution:** Gathering many images of different clothes and giving information about the images and using these images to train a stable diffusion AI model and generating image.
 3. **Target Users:** Fashion designers, filmmakers and costume designers, virtual fashion platforms, artist and graphic designers
 4. **Expected Outcome:** High-Quality, Realistic images, Increased Design Iteration, Reduced Environmental Impact, Enhanced Customer Engagement.
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Phase-2: Requirement Analysis

Objective:

Technical Requirements are software with python based environment, hardware with GPU, OS, storage and networking

Functional Requirements are image generation, image customization, image output, feedback and model update

Key Points:

1. **Technical Requirements:** (python, GPU , OS)
 2. **Functional Requirements:**(input requirements, image generation, customization, output, user interaction)
 3. **Constraints & Challenges**(Technical and data constraints and challenges are ethical and image quality, advancements in AI)
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Phase-3: Project Design

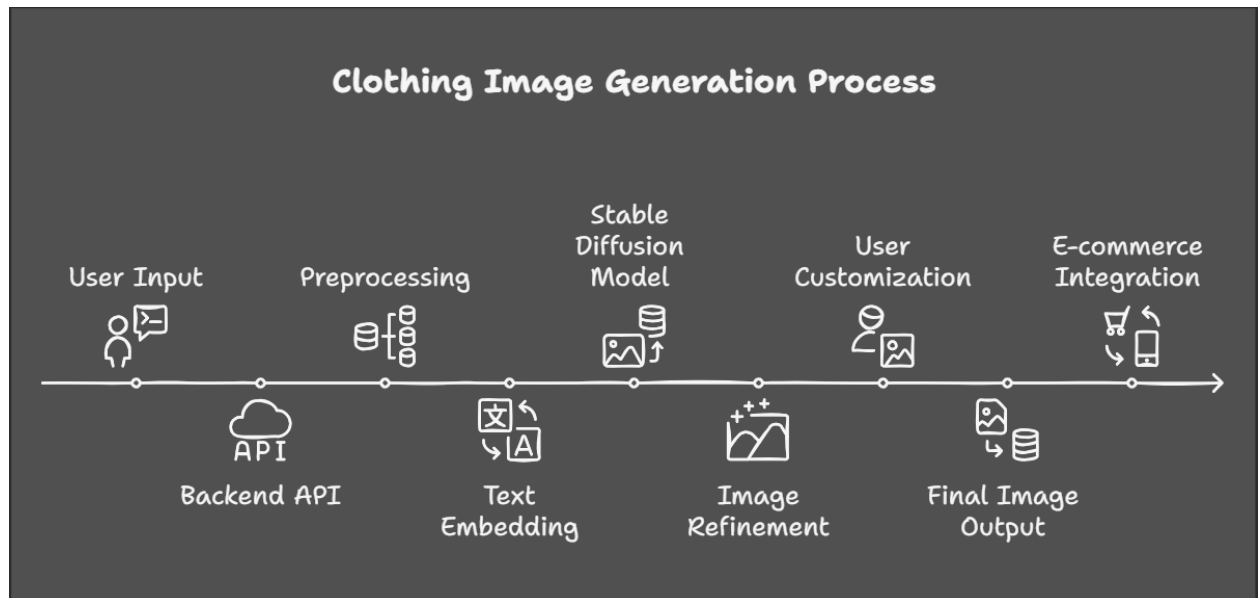
Objective:

The architecture of clothing image generator using stable diffusion pipeline is frontend like html, css , java script and backend like node.js, python, text and coder image generator

The user flow of clothing image generator using stable diffusion pipeline is user input, text encoding, image generation, image processing, image display, model update and user feedback

Key Points:

1. **System Architecture Diagram:**



2. **User Flow:** Our is a open source platform there is no use of login to our page. If you give a prompt as a input we will give the required output.
3. **UI/UX Considerations:** (If applicable, wireframe or basic layout)

Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down the tasks using Agile methodologies.

Key Points:

1. **Sprint Planning:** Our team has divided the task and worked on it. We have assigned the work on different categories. By this planning we can have a better solution to our project.
 2. **Task Allocation:** Our team consists of 5 members We 5 have shared different task and working on it. The team lead is editing the document and 2 of our team members are searching the required matter for documentation and the remaining 2 members are working on the code for project.
 3. **Timeline & Milestones:** We have completed each task in minimum 15-20 mins. We have got the glance on our project. If we are facing any issues we are clearing our issues by taking the help of our mentors.
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Phase-5: Project Development

Objective:

Install dependencies

!pip install diffusers transformers accelerate torch streamlit --quiet

Import libraries

import torch

import streamlit as st

from diffusers import StableDiffusionPipeline

from PIL import Image

Load Stable Diffusion pipeline

pipe = StableDiffusionPipeline.from_pretrained("runwayml/stable-diffusion-v1-5")

pipe.to("cuda" if torch.cuda.is_available() else "cpu")

Streamlit UI

st.title("Clothing Image Generator with Stable Diffusion")

prompt = st.text_input("Enter a clothing description:", "A futuristic streetwear jacket with neon accents, ultra-modern design, high fashion")

if st.button("Generate Image"):

with st.spinner("Generating image..."):

image = pipe(prompt).images[0]

image.save("generated_clothing.png")

st.image(image, caption="Generated Clothing Image",
use_column_width=True)

Key Points:

1. **Technology Stack Used:** Segmind Stable Diffusion API, Stability AI API, Python, GPU.
 2. **Development Process:** Data collection, Data processing, Model training, Model evaluation, Model refining, deployment, testing and maintenance by the following steps we have developed the project.
 3. **Challenges & Fixes:** challenges are low quality images, text to image mismatch, limited diversity.
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Phase-6: Functional & Performance Testing

Objective:

The output is generated (clothing image generator.v1)

Key Points:

1. **Test Cases Executed:** we have given the required prompts all the test cases are executed
 2. **Bug Fixes & Improvements:** we have found the errors in passing the prompts and we have fixed the bugs by that we have got the improved output
 3. **Final Validation:** (Does the project meet the initial requirements?)
 4. **Deployment (if applicable):** (Hosting details or final demo link)
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Final Submission

1. **Project Report Based on the templates**
 2. **Demo Video (3-5 Minutes)**
 3. **GitHub/Code Repository Link**
 4. **Presentation**
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